

② 單元1 坐標系與函數圖形

(主題 14~23：函數及其圖形、一元二次不等式)

答案

1. 37
2. 2
3. 三
4. -3
5. (1, -1)
6. $-\frac{4}{5}$
7. $x \geq -2$
8. 2
9. $x < -\frac{1}{7}$ 或 $x > 7$
10. 無實數解

1. $f(2) = 3 \times 2^3 + 5 = 29$

$f(0) = 8$

故 $f(2) + f(0) = 29 + 8 = 37$

2. $f(-2) = 2 \times (-2)^2 + a \times (-2) + 5 = 13 - 2a$

$f(1) = 2 \times 1^2 + a \times 1 + 5 = 7 + a$

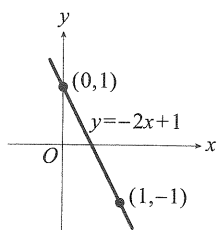
$\therefore f(-2) = f(1)$

$\therefore 13 - 2a = 7 + a \Rightarrow -3a = -6 \Rightarrow a = 2$

3. 令 $y = -2x + 1$,

x	0	1
y	1	-1

其圖形如圖



故此函數的圖形不會經過第三象限

4. $\therefore$ 拋物線通過點 $(-1, 0)$

$\therefore 0 = (-1)^2 + a \times (-1) + b$

$\Rightarrow -a + b = -1 \dots\dots ①$

$\therefore$ 拋物線通過點 $(2, 0)$

$\therefore 0 = 2^2 + a \times 2 + b \Rightarrow 2a + b = -4 \dots\dots ②$

② - ① : $3a = -3 \Rightarrow a = -1$

$a = -1$ 代回 ① : $-(-1) + b = -1 \Rightarrow b = -2$

故 $a + b = -1 + (-2) = -3$

5. $y = 2x^2 - 4x + 1$

$= 2(x^2 - 2x + 1) + 1 - 2 = 2(x - 1)^2 - 1$

故此二次函數的頂點為 $(1, -1)$

6. 令 $a = 5, b = 6, c = 1$

$$\begin{aligned} \text{最小值} &= \frac{4ac - b^2}{4a} = \frac{4 \times 5 \times 1 - 6^2}{4 \times 5} \\ &= \frac{20 - 36}{20} = \frac{-16}{20} = -\frac{4}{5} \end{aligned}$$

7. 原式 : $-2x - 14 \leq 4x - 2$

$\Rightarrow -2x - 4x \leq -2 + 14$

$\Rightarrow -6x \leq 12 \xrightarrow{+(-6)} x \geq -2$

8. 因式分解 $x^2 - 2x - 3 = (x - 3)(x + 1)$

令 $(x - 3)(x + 1) = 0 \Rightarrow x = 3$ 或 -1

則 $x^2 - 2x - 3 < 0$ 的解為 $-1 < x < 3$

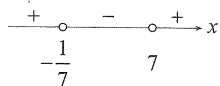
故 $a = -1, b = 3$, 因此 $a + b = -1 + 3 = 2$



9. 因式分解 $7x^2 - 48x - 7 = (7x + 1)(x - 7)$

令 $(7x + 1)(x - 7) = 0 \Rightarrow x = -\frac{1}{7}$ 或 7

則 $7x^2 - 48x - 7 > 0$ 的解為 $x < -\frac{1}{7}$ 或 $x > 7$



10. $\therefore x^2 - 6x + 9 = x^2 - 2 \times 3 \times x + 3^2 = (x - 3)^2$

$\Rightarrow$ 恆正或零

$\therefore x^2 - 6x + 9 < 0 \Rightarrow$ 無實數解

③ 單元2 三角函數

(主題 1~14：有向角、銳角與任意角的三角函數)

答案

1. 丙、丁

2. $\frac{25\pi}{2}$

3. $\frac{43}{20}$

4. $\frac{7}{2}$

5. 1

6. $2\sqrt{2}$

7. 19

8. 甲

9. 三

10. -3

1. 甲 : $-640^\circ = 360^\circ \times (-2) + 80^\circ$

而 $0^\circ < 80^\circ < 90^\circ \Rightarrow$ 甲是第一象限角

乙 : $90^\circ < 123^\circ < 180^\circ \Rightarrow$ 乙是第二象限角

丙 : $270^\circ < 275^\circ < 360^\circ \Rightarrow$ 丙是第四象限角

丁 : $640^\circ = 360^\circ \times 1 + 280^\circ$

而 $270^\circ < 280^\circ < 360^\circ \Rightarrow$ 丁是第四象限角



2. 圓心角 $45^\circ = 45 \times \frac{\pi}{180} = \frac{\pi}{4}$ (弧度)

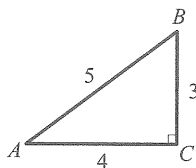
扇形面積 $= \frac{1}{2} \times 10^2 \times \frac{\pi}{4} = \frac{25\pi}{2}$

3. $\triangle ABC$ 的圖形如右

則 $\sin A = \frac{3}{5}$, $\cos A = \frac{4}{5}$,

$\tan A = \frac{3}{4}$

故 $\sin A + \cos A + \tan A = \frac{3}{5} + \frac{4}{5} + \frac{3}{4} = \frac{43}{20}$



4. $a^2 = (\sin 45^\circ)^2 = \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{2}$

$b^2 = (\tan 45^\circ)^2 = 1^2 = 1$

$c^2 = (\sec 45^\circ)^2 = (\sqrt{2})^2 = 2$

故 $a^2 + b^2 + c^2 = \frac{1}{2} + 1 + 2 = \frac{7}{2}$

5. $\sin 75^\circ = \sin(90^\circ - 15^\circ) = \cos 15^\circ$

原點至點 $(\sin 15^\circ, \sin 75^\circ)$ 的距離

$= \sqrt{(\sin 15^\circ - 0)^2 + (\sin 75^\circ - 0)^2}$

$= \sqrt{\sin^2 15^\circ + \sin^2 75^\circ}$

$= \sqrt{\sin^2 15^\circ + \cos^2 15^\circ} = \sqrt{1} = 1$

6. θ 為銳角

以 $\tan \theta = \sqrt{2} = \frac{\sqrt{2}}{1}$ 來作圖

則 $\sin \theta = \frac{\sqrt{2}}{\sqrt{3}}$, $\cos \theta = \frac{1}{\sqrt{3}}$

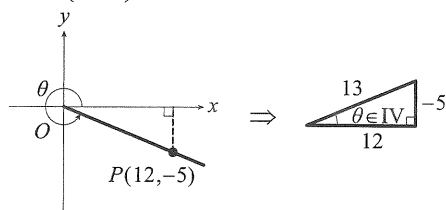
故 $\sqrt{3} \sin \theta + \sqrt{6} \cos \theta = \sqrt{3} \times \frac{\sqrt{2}}{\sqrt{3}} + \sqrt{6} \times \frac{1}{\sqrt{3}}$
 $= \sqrt{2} + \sqrt{2} = 2\sqrt{2}$

7. $\overline{OP} = \sqrt{12^2 + (-5)^2} = \sqrt{169} = 13$

則 $\sin \theta = \frac{-5}{13}$, $\cos \theta = \frac{12}{13}$

故 $13 \sin \theta + 26 \cos \theta$

$= 13 \times \left(-\frac{5}{13}\right) + 26 \times \frac{12}{13} = -5 + 24 = 19$



8. $\therefore 3073^\circ \div 360^\circ = 8 \cdots 193^\circ$

而 $180^\circ < 193^\circ < 270^\circ$

$\therefore \theta = 3073^\circ$ 是第三象限角

則 $\sin \theta < 0$, $\tan \theta > 0$, $\sec \theta < 0$, $\csc \theta < 0$

故甲是正確的

9. $\therefore -400^\circ = 360^\circ \times (-2) + 320^\circ$ 且

$270^\circ < 320^\circ < 360^\circ$

$\therefore -400^\circ$ 是第四象限角 $\Rightarrow \sin(-400^\circ) < 0$

$\therefore 580^\circ = 360^\circ + 220^\circ$ 且 $180^\circ < 220^\circ < 270^\circ$

$\therefore 580^\circ$ 是第三象限角 $\Rightarrow \cos 580^\circ < 0$

故點 $(\sin(-400^\circ), \cos 580^\circ)$ 在第三象限

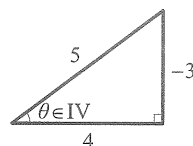
10. $\therefore -90^\circ < \theta < 0^\circ$

$\therefore \theta$ 是第四象限角

以 $\tan \theta = -\frac{3}{4} = \frac{-3}{4}$ 來作圖

則 $\sin \theta = \frac{-3}{5}$, $\cos \theta = \frac{4}{5}$

故 $\frac{\cos \theta + 1}{\sin \theta} = \frac{\frac{4}{5} + 1}{\frac{-3}{5}} = \frac{\frac{9}{5}}{\frac{-3}{5}} = -3$



4 單元 2 三角函數

(主題 15~26：三角函數的轉換與圖形、正餘弦定理)

答案

- | | |
|--------------------------|--------------------|
| 1. 0 | 2. $-\frac{1}{2}$ |
| 3. -1 | 4. 甲 |
| 5. 5π | 6. $3\sqrt{2}$ |
| 7. 2 | 8. $2\sqrt{7}$ |
| 9. $\frac{2\sqrt{5}}{5}$ | 10. $1200\sqrt{5}$ |

1. $\sin \frac{\pi}{3} = \sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos \frac{\pi}{6} = \cos 30^\circ = \frac{\sqrt{3}}{2}$

$\tan \frac{\pi}{4} = \tan 45^\circ = 1$

$\cot\left(-\frac{\pi}{4}\right) = \cot(-45^\circ) = -\cot 45^\circ = -1$

$\sin\left(-\frac{11\pi}{6}\right) = \sin(-330^\circ) = -\sin 330^\circ$

$= -\sin(360^\circ - 30^\circ)$

$= -(-\sin 30^\circ) = \sin 30^\circ = \frac{1}{2}$

$$\cos \frac{\pi}{3} = \cos 60^\circ = \frac{1}{2}$$

$$\begin{aligned}\text{所求} &= \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + 1 \times (-1) + \frac{1}{2} \times \frac{1}{2} \\ &= \frac{3}{4} + (-1) + \frac{1}{4} = 0\end{aligned}$$

$$2. \sin 120^\circ = \sin(180^\circ - 60^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 210^\circ = \cos(180^\circ + 30^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$$

$$\tan 315^\circ = \tan(360^\circ - 45^\circ) = -\tan 45^\circ = -1$$

$$\begin{aligned}\cos(-300^\circ) &= \cos 300^\circ = \cos(360^\circ - 60^\circ) \\ &= \cos 60^\circ = \frac{1}{2}\end{aligned}$$

$$\text{所求} = \frac{\sqrt{3}}{2} + \left(-\frac{\sqrt{3}}{2}\right) + (-1) + \frac{1}{2} = -\frac{1}{2}$$

$$\begin{aligned}3. \text{ 所求} &= \frac{\cos \theta}{\sin \theta} + \frac{\tan \theta}{-\tan \theta} - \frac{-\cos \theta}{-\sin \theta} \\ &= \frac{\cos \theta}{\sin \theta} + (-1) - \frac{\cos \theta}{\sin \theta} = -1\end{aligned}$$

4. 圖形是 $y = \cos x$ 以 x 軸為基準上下拉伸 3 倍
即 $y = 3\cos x$ ，故選甲

5. $\because y = \sin x$ 的週期是 2π
而 $y = \sin 2x$ 的 x 係數是 2

$$\therefore y = \sin 2x \text{ 的週期為 } \frac{2\pi}{2} = \pi, \text{ 即 } a = \pi$$

而 $y = 2\sin x$ 的週期與 $y = \sin x$ 的相等
都是 2π ，即 $b = 2\pi$

$$\text{故 } a + 2b = \pi + 2 \times (2\pi) = \pi + 4\pi = 5\pi$$

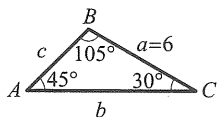
6. $\angle A = 180^\circ - \angle B - \angle C = 180^\circ - 105^\circ - 30^\circ = 45^\circ$
由正弦定理

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

$$\Rightarrow \frac{6}{\sin 45^\circ} = \frac{c}{\sin 30^\circ}$$

$$\Rightarrow \sin 45^\circ \times c = \sin 30^\circ \times 6 \Rightarrow \frac{\sqrt{2}}{2} \times c = \frac{1}{2} \times 6$$

$$\Rightarrow c = \frac{6}{\sqrt{2}} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$



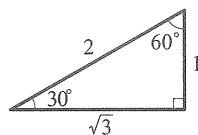
$$7. \because a : b : c = \sin A : \sin B : \sin C = 1 : \sqrt{3} : 2$$

$$\therefore \angle A = 30^\circ, \angle B = 60^\circ, \angle C = 90^\circ$$

$$\text{則 } \sin A = \sin 30^\circ = \frac{1}{2}, \cos B = \cos 60^\circ = \frac{1}{2},$$

$$\sin C = \sin 90^\circ = 1$$

$$\text{故 } \sin A + \cos B + \sin C = \frac{1}{2} + \frac{1}{2} + 1 = 2$$



$$8. \text{ 令 } a = \overline{BC}, b = \overline{AC}, c = \overline{AB}$$

由餘弦定理

$$\begin{aligned}a^2 &= b^2 + c^2 - 2bc \cos A \\ &= 4^2 + 2^2 - 2 \times 4 \times 2 \times \cos 120^\circ\end{aligned}$$

$$= 16 + 4 - 2 \times 4 \times 2 \times \left(-\frac{1}{2}\right) = 28$$

$$\text{則 } a = \sqrt{28} = 2\sqrt{7}, \text{ 故 } \overline{BC} = 2\sqrt{7}$$

$$9. \overline{AD} = \sqrt{3^2 + 1^2} = \sqrt{10}, \overline{AB} = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$

由餘弦定理，在 $\triangle ABD$ 之中

$$\begin{aligned}\cos \theta &= \frac{\overline{AB}^2 + \overline{AD}^2 - \overline{BD}^2}{2 \times \overline{AB} \times \overline{AD}} \\ &= \frac{(3\sqrt{2})^2 + (\sqrt{10})^2 - 2^2}{2 \times 3\sqrt{2} \times \sqrt{10}} = \frac{24}{12\sqrt{5}} \\ &= \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}\end{aligned}$$

$$10. \text{ 令 } s = \frac{1}{2} \times (70 + 80 + 90) = 120$$

由海龍公式

$$\begin{aligned}\text{面積} &= \sqrt{120 \times (120 - 70) \times (120 - 80) \times (120 - 90)} \\ &= \sqrt{120 \times 50 \times 40 \times 30} = 1200\sqrt{5}\end{aligned}$$

⑤ 單元 3 平面向量

(主題 1~8：向量及其基本運算)

答案

1. 4

2. 8

3. (5, -5)

4. -3

5. (-2, 3)

6. 3

7. 2

8. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

9. $\left(\frac{3}{5}, -\frac{4}{5}\right)$

10. -4